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Specialty: Artificial Intelligence and Data Sciences

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Abstract

Semen analysis plays a critical role in the assessment of male fertility, with traditional methods focusing on evaluating parameters such as sperm count, motility, morphology, and DNA integrity. Despite their widespread use, conventional techniques remain time-consuming, labor-intensive, and prone to human error, highlighting the need for automation and precision. The introduction of Computer-Aided Sperm Analysis (CASA) systems has addressed some of these challenges, yet limitations persist in standardization, reproducibility, and scalability.

Recent advances in Artificial Intelligence (AI) offer transformative potential across various domains of sperm analysis. This report explores the integration of Machine Learning (ML), Deep Learning (DL), Computer Vision, Natural Language Processing (NLP), and Ensemble Methods in enhancing the accuracy, efficiency, and reliability of semen evaluation. It delves into specific AI applications such as sperm motility and morphology classification, concentration estimation, DNA fragmentation analysis, predictive modeling, and laboratory automation. The report also reviews publicly available datasets and specialized tools like DeepSperm and SpermQ, which have facilitated AI research in this field.

While AI models demonstrate superior performance metrics compared to traditional methods, several challenges remain, including data scarcity, generalizability across diverse populations, ethical concerns, and clinical integration barriers. The report highlights recent breakthroughs, including the adoption of state-of-the-art architectures like YOLO, ResNet, EfficientNet, and transformer-based models, as well as innovations in generative data augmentation and multi-modal learning.

Through a comparative analysis of AI and traditional techniques, this report evaluates the strengths and weaknesses of both approaches. It concludes by outlining future directions such as explainable AI (XAI), integration with IoT and robotics, and the pursuit of regulatory pathways for clinical adoption. This comprehensive overview aims to guide researchers, clinicians, and developers in advancing AI-powered solutions for more effective and accessible semen analysis.

Keywords: Semen Analysis, Artificial Intelligence (AI), Machine Learning (ML), Deep Learning (DL), Computer Vision, Sperm Morphology, Sperm Motility, DNA Fragmentation, CASA, YOLO, ResNet, EfficientNet, Transformer Models, Generative AI, Multi-modal AI, Explainable AI (XAI), Medical Imaging, Fertility Assessment, Clinical Integration

Dedication

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List of Acronyms :

AI Artificial Intelligence

ANNs Artificial Neural Networks

AUC Area Under the Curve

ARTs Assisted Reproductive Technologies

BMI Body Mass Index

BoF Bag of Freebies

BoS Bag of Specials

CASA Computer-Assisted Sperm Analysis

CNN Convolutional Neural Network

C2PSA Convolutional Parallel Spatial Attention

DFI DNA Fragmentation Index

DNNs Deep Neural Networks

DNA Deoxyribonucleic Acid

E-ELAN Extended Efficient Layer Aggregation Network

FPN Feature Pyramid Network

FPS Frames Per Second

FR-CNN Faster Region-based Convolutional Neural Network

GANs Generative Adversarial Networks

GBT Gradient Boosted Trees

GELAN Generalized Efficient Layer Aggregation Network

GPU Graphics Processing Unit

HuSHeM Human Sperm Head Morphology

IoT Internet of Things

IVF In Vitro Fertilization

LSTM Long Short-Term Memory

MAE Mean Absolute Error

mAP Mean Average Precision

ML Machine Learning

NLP Natural Language Processing

NMS Non-Maximum Suppression

NOA Non-Obstructive Azoospermia

OBB Oriented Bounding Box

R ELAN Residual Efficient Layer Aggregation Network

ResNet Residual Network

RNNs Recurrent Neural Networks

SCIAN-MorphoSpermGS Scientific Analysis of Morphology for Sperm Grading System

SVM Support Vector Machine

SVIA Sperm Video Image Analysis

TAL Task-Aligned Learning

THMA Tail-to-Head Movement Algorithm

U-Net U-shaped Convolutional Network

ViTs Vision Transformers

WISEM Video-based Sperm Evaluation Model

VAP Average Path Velocity

VCL Curvilinear Velocity

VGG Visual Geometry Group

VSL Straight Line Velocity

WHO World Health Organization

XAI Explainable Artificial Intelligence

YOLO You Only Look Once

1.Introduction

For centuries, microscopic imaging has been a vital component of biologic and medical research, offering important insights into the structure, function, and anomalies of cells. Microscopic photos are essential for identifying and examining fertility problems in reproductive medicine. One of the most important diagnostic techniques for assessing male fertility is sperm analysis.

Sperm analysis is a procedure used to assess the health of male reproduction by determining factors such as concentration, mobility, and structure. It is essential for identifying infertility, directing ARTs like in vitro fertilization (IVF), and assessing the effectiveness of medical or surgical treatments.

The primary method of sperm analysis is manual microscopy, in which highly qualified laboratory technical personnel use a micro-scoop to visually evaluate sperm. Despite being the gold standard for many years, this process has a number of drawbacks. Among them are:

- **Subjectivity:** Human inconsistency in interpreting results.
- **Time Consumption:** Labor-intensive processes that take time.
- **Inter-Observer Variability:** Inconsistency in results between observers.
- **Limited Scalability:** Inefficiency in processing large batches of samples.

Computer-Assisted Sperm Analysis (CASA) systems were created in order to overcome these drawbacks. Using imaging software, CASA systems automate certain aspects of sperm analysis, including concentration counts and motility measurements. However, CASA systems are costly to set up, need specialized training, and are unable to reliably determine complex parameters like morphology and DNA fragmentation.

Sperm analysis was transformed by the introduction of Artificial Intelligence (AI) in the form of machine learning (ML) and deep learning (DL). AI-based systems are capable of processing microscopical images with a level of accuracy, speed, and consistency that surpasses human capabilities, AI is capable of:

- Automating the recognition of sperm cells and their classification.
- Providing objective and reproducible results.
- Analyzing complex parameters such as morphology and motility in finer detail.

- Integrating into existing laboratory workflows to enhance efficiency.

The application of AI in sperm analysis is more profound than automation. AI can transform diagnosis of fertility to enable it to:

- Early detection of subtle abnormalities that would be missed using existing methods.
- Creation of personalized treatment strategies based on in-depth analysis of sperm parameters.
- Performance of large-scale studies to generate new knowledge on male reproductive health.

As the demand for high-quality, efficient, and accurate diagnosis of fertility grows, AI's entry in sperm analysis is a milestone in that aspect.

2. Traditional Methods of Sperm Analysis

2.1 Overview of Traditional Sperm Analysis

In traditional sperm analysis, skilled technicians or embryologists manually examine sperm samples under a microscope. For many years, this technique has been the gold standard for evaluating important sperm quality parameters, such as : [\[1\]](#)

- **Sperm Concentration:** The number of sperm cells per unit volume of semen [\[1\]](#)
- **Sperm Motility:** The ability of sperm to move, classified into categories such as rapid progressive, slow progressive, non-progressive, and immotile. [\[1\]](#)
- **Sperm Morphology:** The shape and structure of sperm cells, which can indicate abnormalities. [\[1\]](#)

In order to categorize sperm motility according to their movement patterns, the World Health Organization (WHO) has set specific criteria. For instance, the table below illustrates how sperm are divided into various grades based on their velocity and direction of motion: [\[1\]](#)

No.	Grade	Name	Movements ($\mu\text{m/s}$)
1	A	Rapid progressive motility	above 25
2	B	Slow progressive motility	5–25
3	C	Non-progressive motility	0–5
4	D	Immotility	0

Table 1 : Human Sperm Motility Classification by World Health Organization (WHO) [\[1\]](#)

2.2 Steps in Traditional Sperm Analysis

The traditional sperm analysis process typically involves the following steps :

1. **Sample Preparation:** The semen sample is collected and prepared for analysis. This may involve diluting the sample to make it easier to examine under a microscope. [\[1\]](#)

2. **Microscopic Examination:** A technician examines the sample under a microscope, manually counting sperm cells and assessing their motility and morphology. [\[1\]](#)

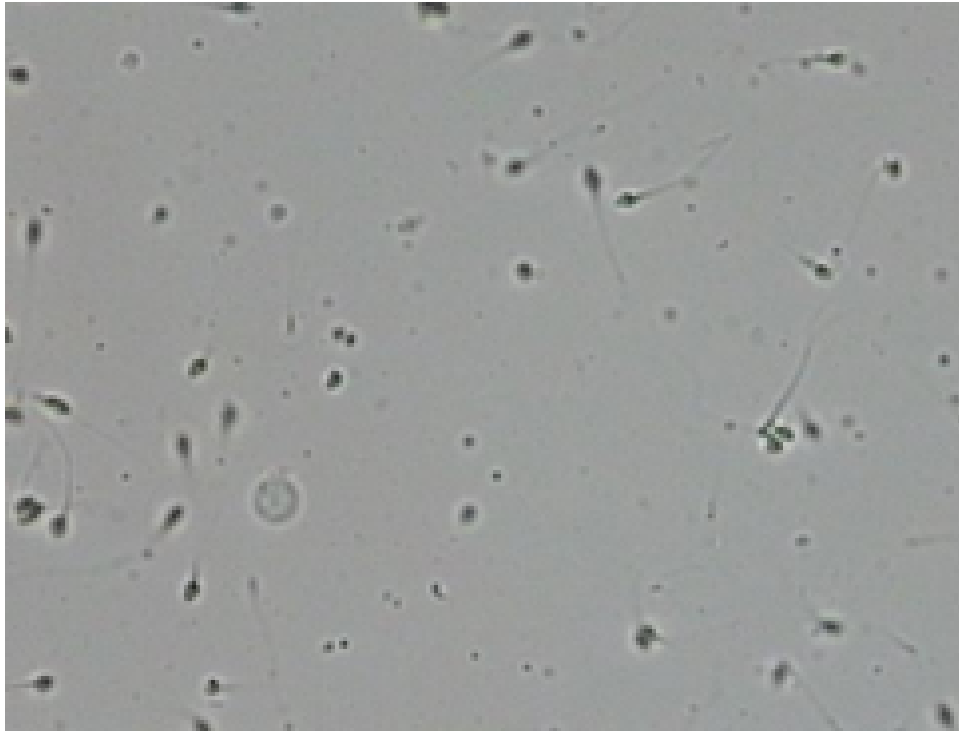


Figure 1 : An example of a microscopic video of sperm. [\[1\]](#)

3. **Classification:** Sperm are classified into different categories based on their movement patterns and shape, following WHO guidelines. [\[1\]](#)
4. **Recording Results:** The technician records the results, including sperm concentration, motility, and morphology, in a report. [\[1\]](#)

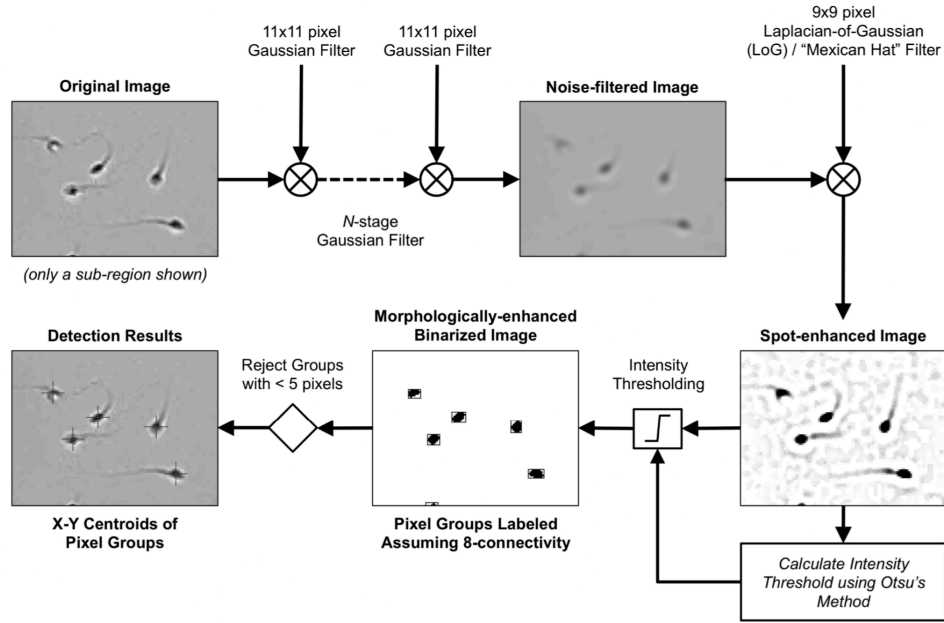


Figure 2: Steps in traditional sperm analysis [1]

2.3 Limitations of Traditional Methods

While traditional sperm analysis has been widely used, it has several significant limitations :

- **Subjectivity:** The results depend heavily on the expertise and judgment of the technician performing the analysis. Different technicians may interpret the same sample differently, leading to variability in results. [1]
- **Time-Consuming:** Manual analysis is labor-intensive and time-consuming, especially when dealing with large numbers of sperm cells.
- **Low Accuracy:** Human error, such as miscounting or misclassifying sperm, can affect the accuracy of the results. [1]
- **Limited Detail:** Traditional methods are less effective at measuring complex parameters such as sperm velocity, linearity, and lateral head displacement, which are important for a comprehensive assessment of sperm quality. [1]
- **Inconsistency:** Without standardized protocols, results may vary between different laboratories and clinics, making it difficult to compare findings. [1]

2.4 The Role of CASA in Addressing Limitations

To address the limitations of traditional methods, **Computer-Assisted Sperm Analysis (CASA)** systems were developed. CASA systems automate the process of sperm analysis, providing more objective, accurate, and efficient results. However, even CASA systems have their own challenges, such as high costs and complexity, which have led to the exploration of advanced solutions like Artificial Intelligence (AI) . [\[1\]](#)

2.5 Summary

Traditional sperm analysis, while widely used, is limited by its subjectivity, time-consuming nature, and potential for human error. These limitations have driven the development of more advanced methods, such as CASA and AI-based systems, which offer greater accuracy, efficiency, and standardization. By transitioning from manual methods to automated systems, we can improve the diagnosis and treatment of male infertility, ultimately benefiting patients and advancing reproductive medicine.

3. AI Techniques in Sperm Analysis

Artificial Intelligence (AI) has revolutionized sperm analysis by introducing advanced techniques that improve accuracy, efficiency, and objectivity. Below are the key AI techniques used in sperm analysis :

3.1 Machine Learning (ML):

Machine Learning (ML) algorithms are widely used in sperm analysis to classify and cluster sperm based on various parameters. ML techniques can be divided into **supervised** and **unsupervised learning** : [\[2\]\[3\]](#)

- **Supervised Learning for Sperm Classification :**

- Supervised learning algorithms are trained on labeled datasets to classify sperm into categories such as normal vs. abnormal. For example: [\[2\]\[3\]](#)
 - **Support Vector Machines (SVM):** Used to classify sperm head morphology into normal, tapered, pyriform, and amorphous categories with high accuracy [\[2\]\[3\]](#)
 - **Random Forest and Naive Bayes:** Applied to classify sperm based on morphology and motility, achieving high precision and recall [\[2\]\[3\]](#)
- These algorithms are particularly useful for identifying sperm with optimal morphology and motility, which are critical for successful fertilization. [\[2\]](#)

- **Unsupervised Learning for Clustering Sperm Morphology :**

- Unsupervised learning algorithms, such as k-means clustering, are used to group sperm based on similarities in morphology or motility without predefined labels. [\[3\]](#)
- For example, unsupervised learning has been applied to cluster sperm into groups based on head shape, midpiece structure, and tail characteristics, helping embryologists identify patterns that may not be visible to the human eye. [\[3\]](#)

3.2 Deep Learning (DL) :

Deep Learning (DL) techniques, particularly Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), have shown remarkable success in sperm analysis: [\[2\]](#)

- **Convolutional Neural Networks (CNNs) for Image Analysis:**
 - CNNs are used to analyze sperm images and extract features such as head shape, acrosome coverage, and vacuole presence. For example:
 - **U-Net Architecture:** Used for segmenting sperm heads, midpieces, and tails in microscopic images with high accuracy .[\[4\]](#)
 - **ResNet and VGG:** Pre-trained models like ResNet and VGG have been fine-tuned for sperm classification tasks, achieving state-of-the-art performance in identifying normal and abnormal sperm.[\[3\]](#)
- **Recurrent Neural Networks (RNNs) for Motility Analysis:**
 - RNNs are used to analyze sperm motility by processing sequential data, such as video frames of sperm movement. For example:
 - RNNs have been applied to track sperm trajectories and classify motility patterns (e.g., progressive, non-progressive, hyperactivated) with high accuracy.[\[2\]\[3\]](#)
- **Transfer Learning Using Pre-trained Models:**
 - Transfer learning involves using pre-trained models (e.g., ResNet, VGG) and fine-tuning them for specific sperm analysis tasks. This approach reduces the need for large labeled datasets and improves model performance.[\[3\]](#)

3.3 Computer Vision :

Computer vision techniques are essential for processing and analyzing sperm images. Key applications include : [\[2\]\[3\]](#)

- **Image Preprocessing Techniques:**
 - Techniques such as segmentation and noise reduction are used to enhance sperm images before analysis. For example:
 - **Thresholding and Edge Detection :** Used to isolate sperm from the background in microscopic images.[\[3\]\[5\]](#)
 - **Morphological Operations :** Applied to remove noise and improve the clarity of sperm structures .[\[3\]\[5\]](#)
- **Feature Extraction :**

- Computer vision algorithms extract features such as sperm morphology (head shape, midpiece length), motility (velocity, trajectory), and concentration (number of sperm per unit volume) .[\[3\]\[5\]](#)
- These features are used as inputs for ML and DL models to classify and analyze sperm.[\[3\]\[5\]](#)

3.4 Natural Language Processing (NLP) :

While less commonly used in sperm analysis, Natural Language Processing (NLP) can be applied to analyze textual data from medical reports and research papers :[\[3\]](#)

- **Text Analysis:**

- NLP algorithms can extract relevant information from medical records, such as patient history, semen analysis results, and treatment outcomes .[\[3\]](#)
- This information can be used to train AI models or provide insights for personalized fertility treatments[\[3\]](#)

3.5 Ensemble Methods :

Ensemble methods combine multiple models to improve the accuracy and robustness of sperm analysis:

- **Model Fusion :**

- Ensemble techniques, such as bagging and boosting, combine the predictions of multiple ML or DL models to achieve higher accuracy [\[5\]](#). For example:
 - A combination of SVM, Random Forest, and CNNs has been used to classify sperm morphology and motility with improved performance.[\[6\]](#)
- These methods are particularly useful when dealing with complex datasets or when individual models have complementary strengths.

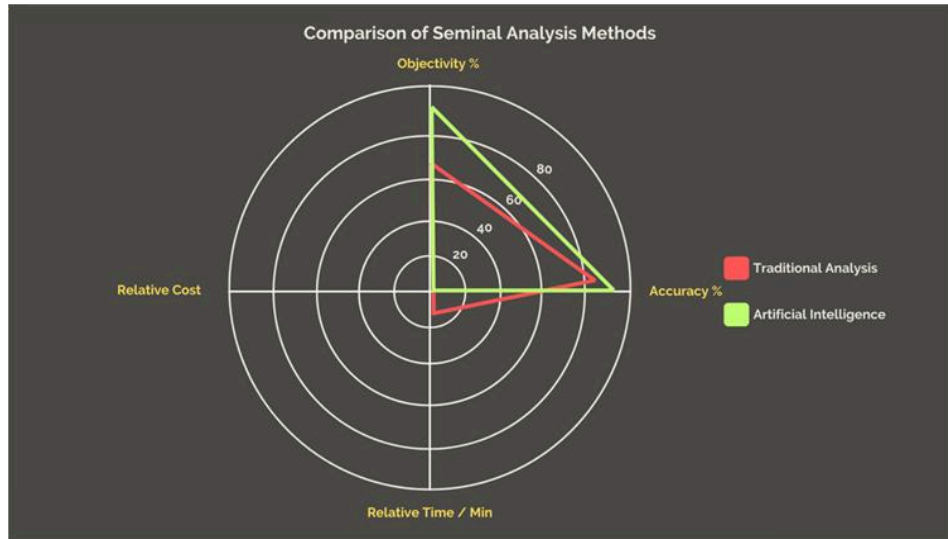


Figure 3: Comparison between traditional and artificial intelligence method in seminal analysis [3]

3.6 Summary

AI techniques, including Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, and Ensemble Methods, have transformed sperm analysis by automating tasks, improving accuracy, and reducing subjectivity. These technologies enable embryologists to analyze sperm morphology, motility, and DNA integrity more efficiently, ultimately improving outcomes in fertility treatments.

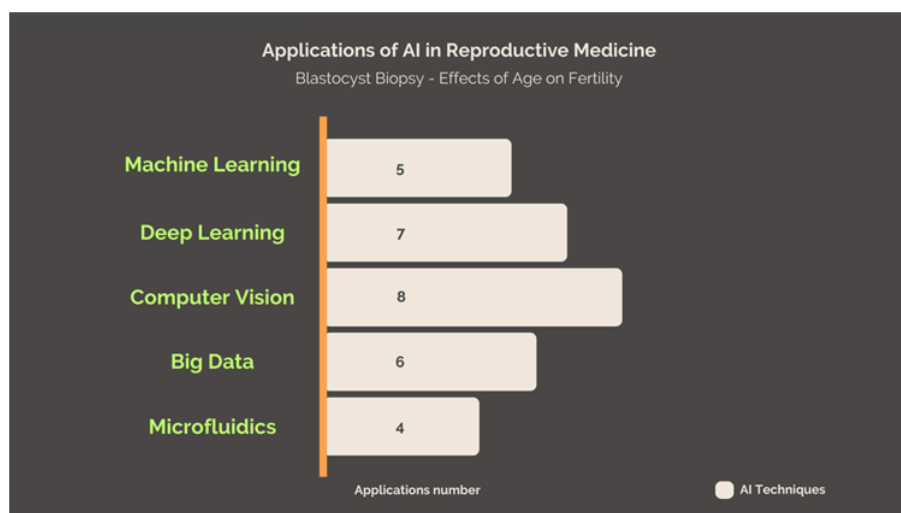


Figure 4: Different AI Techniques with Application Areas in Reproductive Medicine [3]

4. Applications of AI in Sperm Analysis :

Artificial Intelligence (AI) has been increasingly applied in sperm analysis to improve accuracy, efficiency, and objectivity. Below are the key applications of AI in sperm analysis:

4.1 Sperm Motility Analysis

AI is widely used to analyze sperm motility, which is a critical parameter for assessing male fertility. AI algorithms can track and classify sperm movement patterns, providing detailed insights into sperm kinematics.

- **Tracking and Classifying Sperm Movement Patterns:**

- **Convolutional Neural Networks (CNNs):** CNNs are used to analyze video recordings of sperm movement, classifying sperm into categories such as progressive, non-progressive, and immotile. For example, CNNs trained on the VISEM dataset (a multinational video dataset of human spermatozoa) have shown high accuracy in predicting sperm motility [. \[7\]](#)
- **Support Vector Machines (SVM):** SVM models classify sperm motility into categories such as weak, non-vigorous, hyperactivated, intermediate, and progressive, with a predictive accuracy of 89% [. \[7\]](#)
- **Faster Region CNN (FR-CNN):** A novel tail-to-head movement algorithm (THMA) was developed using FR-CNN to classify individual sperm into low, moderate, and high motility with 97% accuracy [. \[7\]](#)

4.2 Sperm Morphology Analysis

AI is used to detect abnormalities in sperm shape and structure, which are critical for assessing sperm quality and fertility potential.

- **Detecting Abnormalities in Sperm Shape and Structure :**

- **Deep Neural Networks (DNNs):** DNNs are trained to classify sperm morphology on head dimensions, acrosome coverage, and vacuole presence. For example, DNNs trained on the Modified Human Sperm Head Morphology Analysis (MHSMA) dataset achieved high accuracy in detecting abnormal sperm [. \[7\]](#)

- **Support Vector Machines (SVM):** SVM models are used to classify sperm heads into normal, tapered, pyriform, and amorphous categories with high precision .[\[7\]](#)

4.3 Sperm Concentration Estimation

AI is used to count sperm cells in a sample, providing a more accurate and efficient alternative to manual counting.

- **Counting Sperm Cells in a Sample :**

- **Artificial Neural Networks (ANNs):** ANNs are used to predict sperm concentration with high accuracy. For example, a full-spectrum neural network (FSNN) model achieved 93% accuracy in predicting sperm concentration .[\[7\]](#)
- **Image Recognition Algorithms:** AI-based image recognition algorithms can estimate sperm concentration and total motile sperm count with high correlation to manual analysis .[\[7\]](#)

4.4 DNA Fragmentation Analysis

AI is used to assess the genetic integrity of sperm, which is critical for predicting fertilization success and embryo development.

- **Assessing Genetic Integrity of Sperm :**

- **Convolutional Neural Networks (CNNs):** CNNs are used to predict DNA fragmentation index (DFI) values based on sperm images. For example, a deep CNN model showed a significant correlation between sperm images and DFI values .[\[7\]](#)
- **Machine Learning Models:** ML models trained on sperm images and corresponding DFI values can predict DNA damage in individual spermatozoa with high accuracy .[\[7\]](#)

4.5 Predictive Modeling

AI is used to predict fertility outcomes based on sperm parameters, helping clinicians make informed decisions.

- **Predicting Fertility Outcomes Based on Sperm Parameters :**

- **Random Forest and Logistic Regression:** These models are used to predict the likelihood of sperm retrieval success in non-obstructive azoospermia (NOA) patients with high sensitivity and specificity.[\[9\]](#)
- **Artificial Neural Networks (ANNs):** ANNs are used to predict the presence of sperm in testicular biopsies with high accuracy, aiding in the decision-making process for surgical sperm retrieval.[\[7\]](#)

4.6 Automation of Laboratory Processes

AI is integrated into laboratory workflows to improve efficiency and reduce human error.

- **Integrating AI into Lab Workflows for Efficiency :**
 - **Computer-Assisted Sperm Analysis (CASA):** AI-enhanced CASA systems automate sperm analysis, reducing the time and effort required for manual assessment .[\[7\]](#)
 - **Smartphone-Based AI Systems:** AI algorithms integrated into smartphone applications allow for rapid and accurate sperm analysis, even in low-resource settings .[\[7\]](#)

4.7 Summary

AI has transformed sperm analysis by automating tasks, improving accuracy, and reducing subjectivity. From tracking sperm motility to predicting fertility outcomes, AI applications are enhancing the efficiency and reliability of sperm analysis in clinical and research settings. By integrating AI into laboratory workflows, clinicians can provide more accurate diagnoses and improve patient outcomes in fertility treatments.

5 Datasets and Tools

5.1 Publicly available datasets:

5.1.1 VISEM Dataset :

The VISEM dataset is a multimodal video dataset designed for AI-based sperm motility analysis. It includes:

- **Video Data:** 85 videos (640×480 pixels at 50 FPS) of human spermatozoa, each 30 seconds long, totaling 29,196 frames. [\[10\]](#)
- **Annotations:** Manually labeled bounding boxes for sperm tracking, with classifications for normal sperm, clusters, and small/pinhead sperm. [\[10\]](#)
- **Clinical Metadata:** Participant data (BMI, abstinence time), semen analysis results (concentration, motility, morphology), and biochemical markers (fatty acids, hormones) . [\[10\]](#)

Key Features:

- **Diverse Samples:** Captures variability in sperm motility and concentration across 20 patients . [\[10\]](#)
- **Benchmark Utility:** Used to train models like YOLOv5, achieving a mean average precision (mAP) of 0.223 for sperm detection . [\[10\]](#)
- **Extension (VISEM-Tracking):** Adds 656,334 annotated objects with tracking IDs, enabling research on sperm kinematics and motility patterns . [\[10\]](#)

Applications:

- Training **deep learning models** for sperm detection and motility classification . [\[10\]](#)
- Validating **real-time tracking algorithms** (e.g., for CASA systems). [\[10\]](#)

5.1.2 SVIA Dataset :

The Sperm Videos and Images Analysis (SVIA) dataset is a comprehensive resource for computer-aided sperm analysis (CASA), featuring:

- **Multimodal Data:**

- 101 microscopic videos (1-3 seconds each) at 30 FPS ..[\[11\]](#)
- 3,622 annotated image frames . .[\[11\]](#)
- 278,000 labeled objects across three subsets:
 - Subset-A: 125,000 detection annotations ..[\[11\]](#)
 - Subset-B: 26,000 segmentation masks ..[\[11\]](#)
 - Subset-C: 125,880 classified objects (sperm vs. impurity) .[\[11\]](#)
- **Key Features:**
 - Focus on tiny object detection (sperm heads average 20×10 pixels) ..[\[11\]](#)
 - Includes challenging cases: low-contrast, overlapping, and fast-moving sperms..[\[11\]](#)
 - Provides motility tracking data for kinematic analysis (VSL, VCL, VAP) ..[\[11\]](#)
- **Benchmark Utility:**
 - Used to train state-of-the-art models like ACTIVE (91.13% AP50 for sperm detection) ..[\[11\]](#)
 - Supports comparative studies of detection architectures (YOLO, Faster R-CNN, SSD)..[\[11\]](#)

Applications:

- Training models for sperm-impurity differentiation . [\[11\]](#)
- Developing real-time motility analysis algorithms .[\[11\]](#)
- Benchmarking tiny object detection in medical imaging .[\[11\]](#)

5.1.3 HuSHeM Dataset

The Human Sperm Head Morphology (HuSHeM) dataset focuses on morphological classification of sperm heads:

- **Data Composition :**
 - 216 high-resolution sperm head images (131×131 pixels) .[\[12\]](#)
 - Fixed and stained semen smears .[\[12\]](#)
 - Four classes: normal, tapered, pyriform, and amorphous .[\[12\]](#)

- **Key Features:**
 - Specialized for head morphology analysis [\[12\]](#)
 - Enables training of models to detect subtle head abnormalities [\[12\]](#)
 - Used in dictionary learning approaches (92% classification accuracy) [\[12\]](#)
- **Applications:**
 - Benchmarking CNN architectures for sperm classification [\[12\]](#)
 - Developing automated morphology scoring systems [\[12\]](#)

5.1.4 SCIAN-MorphoSpermGS Dataset

The SCIAN-MorphoSpermGS dataset extends morphological analysis with finer categorization:

- **Data Composition:**
 - 1,854 annotated sperm images [\[12\]](#)
 - Five classes: normal, tapered, pyriform, small, and amorphous [\[12\]](#)
 - Includes segmentation masks for heads, acrosomes, and nuclei [\[12\]](#)
- **Key Features:**
 - Provides multi-component segmentation (Dice coefficient: 0.90 for heads) [\[12\]](#)
 - Used to train Mask R-CNN models for automated WHO compliance checks [\[12\]](#)
 - Contains challenging cases with overlapping cellular debris [\[12\]](#)
- **Applications:**
 - Training models for strict Kruger morphology assessment [\[12\]](#)
 - Validating AI-powered CASA systems against manual analysis [\[12\]](#)

5.2 Software and Tools:

5.2.1 DeepSperm

DeepSperm is a deep learning framework developed to accurately and efficiently detect sperm cells in densely populated semen videos, particularly in veterinary contexts such as bull sperm

analysis. Designed with the challenges of overlapping cells, low contrast, and rapid motion in mind, DeepSperm delivers robust detection performance in real-time.[\[13\]](#)

- **Key features:**

- **Model Design:** DeepSperm incorporates a streamlined detection layer tailored for identifying small, fast-moving sperm cells. The architecture is optimized for both speed and precision in complex visual environments.[\[13\]](#)
- **Performance:** The model demonstrated a high detection accuracy, achieving a mean Average Precision (mAP) of 86.91% . It also runs at over 50 frames per second, surpassing more general-purpose detectors like YOLOv3 and Mask R-CNN in both accuracy and computational efficiency.[\[13\]](#)
- Addresses challenges like partial occlusion, low contrast, and rapid sperm movement through higher network resolution, dropout layers, and data augmentation.[\[13\]](#)
- Lightweight model with significantly reduced GPU memory and storage requirements compared to traditional methods.[\[13\]](#)

- **Applications:**

- Automated real-time analysis of sperm in high-density samples. [\[13\]](#)
- Suitable for use in AI-based semen evaluation systems in veterinary and agricultural reproductive technologies.[\[13\]](#)

5.2.2 SpermQ

SpermQ is an open-source software tool designed for the quantitative analysis of sperm motility, with a particular focus on the flagellar beat pattern. As an ImageJ plugin, it allows researchers to extract detailed motion parameters from time-lapse microscopy images, facilitating studies on sperm dynamics and steering behavior.[\[14\]](#)

- **Key features:**

- **Detailed Flagellar Analysis:** SpermQ measures key aspects of flagellar movement such as beat frequency, curvature, and amplitude, enabling researchers to gain insights into the biomechanics of sperm motility. [\[14\]](#)

- **Compatibility and Flexibility:** The software works with a variety of imaging methods, including dark-field and fluorescence microscopy, and supports the analysis of sperm from different species.[\[14\]](#)
- **Applications:**
 - Investigating the role of flagellar motion in sperm navigation.[\[14\]](#)
 - Comparative studies on motility across different species or conditions.[\[14\]](#)

6 Challenges and Limitations

6.1 Data Quality and quantity:

A major challenge in developing robust AI systems for sperm analysis lies in the lack of large-scale, well-labeled datasets. Many available datasets are small and often restricted to static images, which are insufficient for tasks requiring temporal information such as sperm motility analysis. Although efforts like the VISEM Tracking dataset offer annotated video data, the field still faces data scarcity due to privacy concerns, ethical limitations, and the labor-intensive nature of medical image annotation.[\[10\]](#)

6.2 Generalization Across Diverse Samples:

AI models trained on datasets from specific populations or labs often struggle to generalize across different sources. This is primarily due to differences in sample preparation, imaging techniques, and inherent biological variability between individuals and populations. Such domain-specific biases hinder the development of models that perform reliably in varied clinical environments.[\[24\]](#)

6.3 Ethical and Privacy Considerations:

The use of AI in sperm analysis raises critical ethical and privacy challenges. These include concerns over data confidentiality, consent for use of reproductive health data, and risks of algorithmic bias when datasets lack diversity. Without proper data governance and ethical oversight, the use of AI could inadvertently reinforce health disparities or misuse sensitive patient information. [\[25\]](#)

6.4 Interpretability of AI Models:

One of the major barriers to clinical acceptance of AI in sperm analysis is the lack of transparency in how models make decisions. Deep learning models, in particular, are often considered "black boxes" because they do not offer explanations for their predictions. This lack of interpretability makes clinicians hesitant to trust or act on AI-generated outputs, especially in critical healthcare settings.[\[26\]](#)[\[27\]](#)

6.5 Integration into Clinical Workflows:

Even when AI models show promising performance in research, integrating them into real clinical workflows remains challenging. This involves overcoming technical issues, validating

models in real-world scenarios, training medical staff, and meeting regulatory standards. Many clinics may also lack the infrastructure needed to deploy AI tools effectively, making seamless adoption difficult without institutional and policy-level support.[\[28\]](#)

7 Recent Advances and Breakthroughs

7.1 State of the art Models:

Recent years have seen major breakthroughs in deep learning, particularly in object detection, classification, and tracking, which are all fundamental for analyzing sperm morphology and motility. Models originally developed for general purpose vision tasks are now being fine-tuned for reproductive health.

7.1.1 YOLO (You Only Look Once)

The YOLO framework has undergone several iterations, each introducing enhancements to improve accuracy, speed, and adaptability.

A. YOLOv2 (2016) - YOLO9000

YOLOv2 introduced significant improvements in training stability and bounding box prediction accuracy. The integration of Batch Normalization helped stabilize and accelerate the training process, while the adoption of Anchor Boxes aligned the model with techniques used in Faster R-CNN, enabling more effective bounding box predictions. Moreover, YOLO9000 extended the model's capabilities to recognize over 9,000 object categories by combining supervised and weakly supervised datasets.[\[17\]](#)

B. YOLOv3 (2018)

YOLOv3 enhanced object detection performance across various object sizes by integrating a Feature Pyramid Network (FPN), allowing for more accurate multi-scale detection. The model also introduced the Darknet-53 backbone, a deeper and more robust feature extractor, which improved both classification and detection accuracy.[\[17\]](#)

C. YOLOv4 (2020)

YOLOv4 focused on performance optimization by integrating Cross-Stage Partial (CSP) connections to reduce computational complexity without sacrificing accuracy. It also introduced the concepts of Bag of Freebies (BoF) and Bag of Specials (BoS), which provided performance enhancements during training and inference without additional computational cost.[\[17\]](#)

D. YOLOv5 (2020)

YOLOv5 marked a shift by being implemented in PyTorch, which facilitated easier adoption and customization by the research community. It emphasized deployment by providing pre-trained models and a user-friendly interface, streamlining the application of object detection in real-world scenarios.[\[17\]](#)

E. YOLOv6 (2022)

YOLOv6 brought a set of innovations tailored for efficiency and deployment. It employed RepVGG-inspired reparameterization techniques to improve speed and accuracy, especially for lightweight networks. Self-distillation was used across classification and regression tasks to enhance performance without increasing inference costs. Additional techniques such as Task-Aligned Learning (TAL), quantization optimization with RepOptimizer, and channel-wise distillation addressed the challenges of label assignment and quantization. An anchor-free design simplified decoding and improved generalization, while the architecture was tailored for industrial applications, especially on low-power GPUs like the Tesla T4.[\[17\]](#)

F. YOLOv7 (2022):

YOLOv7 introduced a range of innovations aimed at maximizing detection accuracy while maintaining real-time performance. The Trainable Bag-of-Freebies framework integrated trainable techniques such as re-parameterization and dynamic label assignment. The Extended-ELAN (E-ELAN) architecture used group convolutions to expand learning capacity without disrupting gradient flow. A planned re-parameterization strategy improved residual-based networks, while coarse-to-fine label assignment boosted detection precision. Additionally, a compound scaling method enabled efficient scaling of concatenation-based models. YOLOv7 achieved state-of-the-art real-time performance, surpassing both transformer-based and CNN-based detectors.[\[17\]](#)

G. YOLOv8 (2023):

YOLOv8 continued the trend of superior real-time detection by achieving the highest accuracy (mAP) and speed (FPS) compared to previous YOLO models and other state-of-the-art detectors like Faster R-CNN and DETR. It introduced a unified framework that supported multiple tasks, such as object detection, instance segmentation, and classification. With an improved architecture featuring enhanced backbone and neck designs, YOLOv8 offered better feature extraction and fusion. The

model adopted an anchor-free detection head and incorporated dynamic label assignment, advanced data augmentation, and optimized training strategies. It also provided an accessible API, a variety of pre-trained models (YOLOv8n to YOLOv8x), and compatibility with deployment frameworks such as TensorRT, ONNX, CoreML, and OpenVINO. Maintained by Ultralytics, YOLOv8 is open-source and community-driven.[\[17\]](#)

H. YOLO-NAS (2023):

YOLO-NAS introduced the use of Neural Architecture Search (NAS) to automatically design an optimized model architecture, enhancing real-time detection performance. It featured a quantization-friendly design that maintained high performance with minimal loss when quantized, making it ideal for deployment on a range of hardware platforms. YOLO-NAS surpassed earlier YOLO versions in terms of mean Average Precision (mAP) and latency, striking a balance between accuracy and computational efficiency. It was released in three variants Small, Medium, and Large each optimized for different hardware and application constraints.[\[17\]](#)

I. YOLOv9 (2024):

YOLOv9 introduced several advanced features to boost detection accuracy and efficiency. It implemented Programmable Gradient Information (PGI), an auxiliary reversible branch designed to preserve essential data across layers, improving gradient generation and minimizing information loss. The model leveraged the Generalized Efficient Layer Aggregation Network (GELAN), which combined the strengths of CSPNet and ELAN to maximize parameter efficiency and speed. YOLOv9 came in multiple variants YOLOv9t, YOLOv9s, YOLOv9m, YOLOv9c, and YOLOv9e catering to diverse needs in terms of speed and computational demands. It delivered superior mAP with fewer parameters and lower computational requirements, making it suitable for a variety of real-time tasks, including object detection and instance segmentation. The model also featured streamlined training procedures and broad compatibility with deployment frameworks like TensorRT and ONNX.[\[15\]](#)

J. YOLOv10 (2024):

YOLOv10 introduces a groundbreaking innovation by eliminating the traditional Non-Maximum Suppression (NMS) step. This streamlines the object detection pipeline, reducing computational overhead and enhancing real-time performance. The

model comes in six distinct variants—YOLOv10-N, YOLOv10-S, YOLOv10-M, YOLOv10-B, YOLOv10-L, and YOLOv10-X—each designed to balance latency and accuracy depending on specific deployment needs. With a mean Average Precision (mAP) reaching up to 54.4% and reduced latency, YOLOv10 demonstrates a significant leap forward in object detection efficiency and performance compared to its predecessors.[\[16\]](#)

K. YOLOv11 (2024):

YOLOv11 brings improvements in detection accuracy through advanced attention mechanisms. It introduces the C2PSA (Convolutional Parallel Spatial Attention) block, which helps the model focus more effectively on critical regions of the image, especially in scenarios with occluded objects. Architecturally, it replaces older C2f blocks with optimized C3k2 modules that utilize smaller kernels ($k=2$), delivering faster computation without sacrificing feature extraction quality. YOLOv11 achieves higher mAP scores than YOLOv8 while using 22% fewer parameters in comparable variants, such as YOLOv11m versus YOLOv8m. Additionally, it expands capabilities to multi-task scenarios like oriented bounding box (OBB) detection and pixel-level instance segmentation. [\[18\]](#)

L. YOLOv12 (2025):

YOLOv12 builds on architectural and computational efficiency with its Segmented Attention Mechanism. This system partitions the spatial domain of feature maps into distinct regions, allowing for more targeted attention while integrating FlashAttention to significantly reduce memory bandwidth demands. The model retains broad receptive fields with minimal overhead. Its backbone incorporates a Residual Efficient Layer Aggregation Network (R-ELAN), designed to improve gradient flow and promote feature reuse across different network depths. Standard convolutions are replaced with 7×7 separable convolutions to preserve spatial context and lower the parameter count, eliminating the need for explicit positional encoding.[\[20\]](#)

M. YOLO-World (2025):

YOLO-World marks a shift toward open-vocabulary object detection, allowing the model to recognize objects outside of predefined categories in real time, without retraining. Operating at 52 FPS on a V100 GPU while maintaining a 35.4 AP on the LVIS benchmark, it leverages a CLIP-based text encoder to seamlessly fuse visual and

linguistic features. This empowers YOLO-World to dynamically interpret user-defined prompts such as categories or descriptive phrases. Its zero-shot capabilities stem from pre-training on diverse datasets like Objects365 and GQA, alongside pseudo-labeled image-text pairs, enabling detection of unseen objects across over 1,200 LVIS categories. The prompt-then-detect paradigm facilitates deployment on edge devices by re-parameterizing vocabulary embeddings into model weights, making it ideal for domains such as autonomous driving, surveillance, and interactive AI, where adaptability to new object classes is critical.[\[21\]](#)

7.1.2 ResNet & EfficientNet

ResNet and EfficientNet are among the most effective convolutional networks for image-based classification tasks, including the categorization of human sperm head morphology. EfficientNet's compound scaling strategy optimizes accuracy and efficiency by adjusting model depth, width, and input resolution in a balanced manner. These networks are particularly suitable for medical datasets, which are often limited in size and contain imbalances, as seen in reproductive health applications. Their proven transferability and high performance make them ideal candidates for clinical deployment in fertility diagnostics. [\[22\]](#)

7.1.3 Transformer Based Vision Models

Although not yet widely implemented in sperm analysis, Vision Transformers (ViTs) and related models are gaining attention for their ability to capture long range spatial dependencies in images. Their use of self-attention mechanisms enables better modeling of global patterns and subtle morphological variations, offering potential improvements over conventional convolutional methods in detecting nuanced abnormalities. Their attention mechanisms are particularly helpful for complex patterns not easily detected by CNNs.[\[22\]](#)

7.1.4 Hybrid & Ensemble models

Recent advancements in sperm analysis leverage the strengths of multiple deep learning architectures by integrating them into ensemble frameworks. These systems typically combine CNNs such as ResNet, VGG, or DenseNet for feature extraction with a meta-classifier that refines the final prediction. This ensemble approach enhances classification accuracy and consistency. While temporal modeling with LSTMs or transformers is more common in motility analysis, ensemble strategies significantly boost the robustness of morphology assessment by reducing misclassifications and improving generalizability across datasets.[\[23\]](#)

Ensemble models combining different AI architectures can enhance robustness and reduce false positives.[\[23\]](#)

7.2 Novel Techniques:

7.2.1 Generative AI for Data Augmentation

Generative Adversarial Networks (GANs) are capable of generating synthetic sperm images and enhancing the quality of images captured under suboptimal conditions. These techniques expand training datasets for classification algorithms, reducing overfitting and improving model robustness in heterogeneous clinical environments. GANs consist of two competing networks (generator and discriminator) that create and validate synthetic images, enabling the generation of realistic sperm images for AI training . [\[3\]](#)

7.2.2 Multi-modal AI combining image and clinical data

AI integrates multi-omics data (genomic, epigenetic, proteomic) with traditional semen parameters (motility, morphology) to build comprehensive predictive models. For example, algorithms correlate genetic variants or abnormal methylation levels with infertility risk. AI-driven models combine male factors (semen quality, genomics), female factors (age, ovarian reserve), and embryonic data to predict ART success rates .

7.3 Specialized Sperm Analysis Models

7.3.1 YOLOv8E TrackEVD framework

Recent research has introduced innovative deep learning approaches specifically designed for semen analysis. One notable advancement is the Sperm YOLOv8E TrackEVD framework, which combines enhanced detection and tracking capabilities for automated sperm evaluation.[\[19\]](#)

The detection component (Sperm YOLOv8-E) incorporates several architectural improvements to address the challenges of sperm analysis: [\[19\]](#)

- An additional high-resolution (160×160) detection layer specifically designed for small sperm objects.[\[19\]](#)
- Two complementary attention mechanisms (SegNext Attention and EMA) to better capture sperm features.[\[19\]](#)
- SPDConv modules to preserve fine details during feature extraction .[\[19\]](#)

- A novel combination of Inner-MPDIoU and NWD loss functions optimized for small object detection.[\[19\]](#)

For sperm tracking, the framework employs an improved DeepOCSORT algorithm (SpermTrack-EVD) featuring:

- OSNet-based re-identification for better feature representation.[\[19\]](#)
- Enhanced appearance matching capabilities.[\[19\]](#)
- Camera motion compensation.[\[19\]](#)

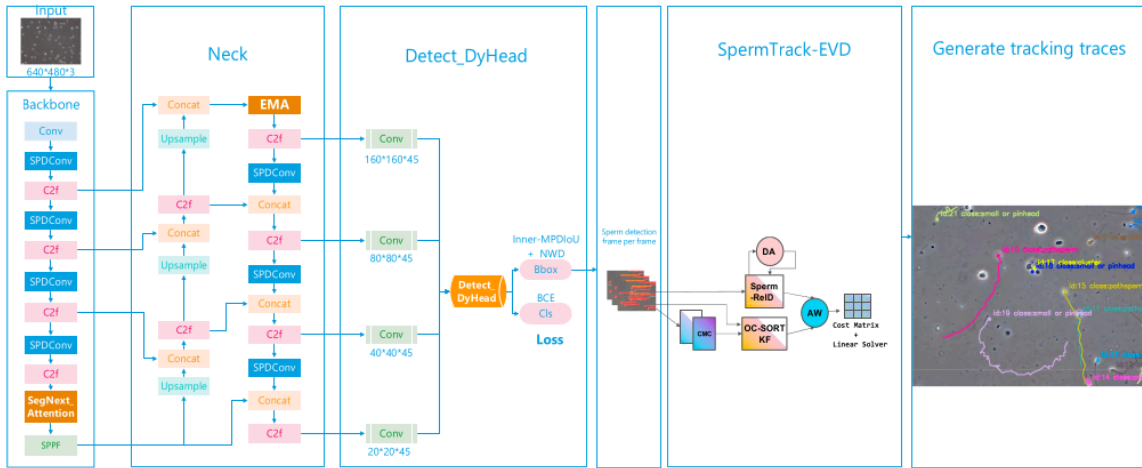


Figure 5: The architecture of YOLOv8E-TrackEVD [\[19\]](#)

7.3.1 RetinaNet and Modified CSR-DCF for Sperm Detection and Tracking

Recent advancements in sperm motility analysis have leveraged the strengths of deep learning and object tracking to improve precision in both detection and temporal localization. A noteworthy contribution is the integration of a RetinaNet-based detection model with a customized tracking method known as Modified CSR-DCF (Discriminative Correlation Filter with Channel and Spatial Reliability).[\[29\]](#)

In this approach, sperm cells are initially localized using RetinaNet, a one-stage object detector known for its balance between accuracy and speed. To enhance robustness, the

detection pipeline processes a sequence of concatenated video frames rather than relying solely on single-frame analysis. This multi-frame strategy provides contextual temporal information, which helps in minimizing false positives and capturing transient appearances of sperm cells in noisy phase-contrast microscopy videos.[\[29\]](#)

Following detection, the tracking component employs a modified CSR-DCF algorithm, fine-tuned to handle the unique motion patterns of sperm cells, such as abrupt directional changes and intermittent disappearance due to optical noise. The modification involves introducing a multi-scale tracking strategy that adapts dynamically to sperm size and velocity variations during video progression.[\[29\]](#)

Quantitatively, this hybrid system achieved a detection average precision (AP) of 99.1% and a tracking F1-score of 96.61%, clearly outperforming previous benchmarks including HDBN and baseline CSR-DCF trackers. These results underscore the effectiveness of integrating deep feature-based detection with biologically informed motion modeling for sperm tracking in real-time analysis scenarios.[\[29\]](#)

This method exemplifies a significant advance in AI-assisted sperm analysis, particularly for motility studies, and highlights the potential of combining vision transformers and spatially aware trackers in future research.[\[29\]](#)

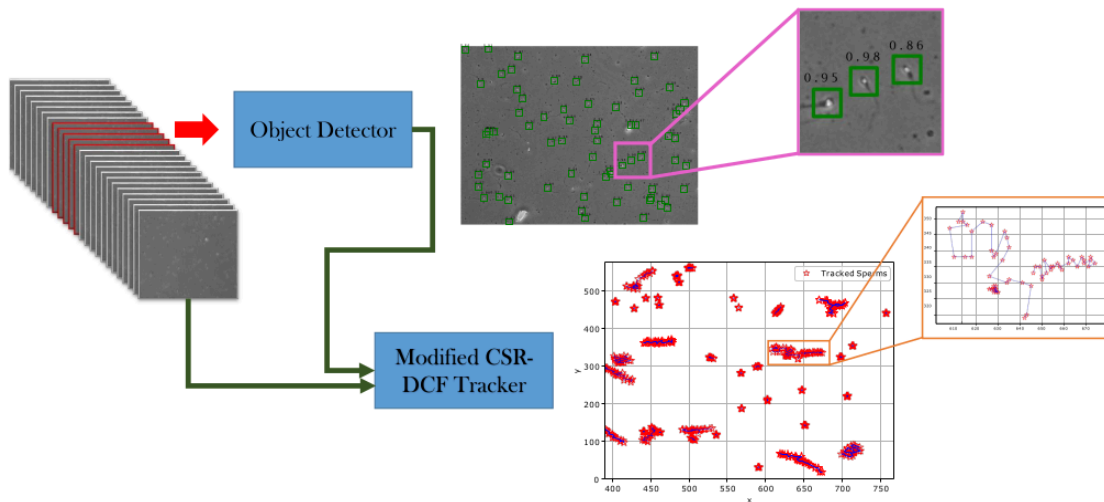


Figure 6: Workflow of the proposed method: concatenated video frames are used for detection, followed by tracking with the modified CSR-DCF algorithm.[\[29\]](#)

8 Comparative Analysis of AI and Traditional Methods in Semen Analysis

Semen analysis is a cornerstone of male infertility diagnosis, but traditional manual methods face challenges in standardization and accuracy. Below is a comparative synthesis of AI-driven approaches versus traditional techniques, performance of AI models, and commercial solutions, supported by recent scientific research and market insights

8.1 AI vs. Traditional Methods

8.1.1 Advantages of AI

- **Standardization & Precision:** Automated systems like CASA (Computer Assisted Semen Analysis) reduce human error and variability. A double blind study comparing SQA-V GOLD (AI-based) and manual methods found strong correlations (Spearman $\rho = 0.93 - 0.96$) for sperm concentration and motility, with AI offering higher precision. [\[30\]](#)
- **Speed & Efficiency:** AI processes samples in minutes, while manual analysis can take hours. For example, Microptic's CASA system analyzes sperm concentration and motility in < 1 minute. [\[7\]](#)[\[31\]](#)
- **Complex Pattern Recognition:** AI detects subtle sperm abnormalities (e.g., DNA fragmentation) with > 90% accuracy, surpassing human visual assessment. [\[25\]](#) [\[7\]](#)
- **Scalability:** AI integrates with telehealth platforms, enabling remote analysis and home testing kits (e.g., ExSeed Health's smartphone-based device). [\[31\]](#) [\[32\]](#)

8.1.2 Disadvantages of AI

- **Morphology Limitations:** Automated systems struggle with atypical samples (e.g., debris-heavy semen) and lack nuanced morphology assessment, requiring manual validation. [\[30\]](#)
- **Cost & Accessibility:** High-end AI systems (e.g., Hamilton Thorne CEROS) are expensive, limiting adoption in low-resource regions. [\[31\]](#)[\[33\]](#)

8.1.3 Traditional Methods

- **Human Expertise:** Manual microscopy allows contextual interpretation of morphology but suffers from inter-observer variability (e.g., 15–20% discrepancy in motility assessment). [\[25\]](#) [\[7\]](#)
- **Flexibility:** Adaptable to small datasets but inefficient for large-scale analysis.[\[33\]](#) [\[30\]](#)

8.2 Performance of AI Models in Semen Analysis

Model/Technique	Application	Performance
Support Vector Machine (SVM)	Motility classification	89.9% accuracy on 2,817 sperm samples
Deep Neural Networks	Morphology assessment	AUC 88.59% for abnormal sperm detection
Random Forest	IVF outcome prediction	AUC 84.23% for predicting fertilization success
Convolutional Neural Networks	Motility tracking	Mean absolute error (MAE) of 2.92 for motility prediction

Table 2: AI models vary in accuracy, speed, and specialization.[\[25\]](#) [\[7\]](#)

- **Clinical Validation:** AI models like gradient boosting trees (GBT) predict sperm retrieval success in non-obstructive azoospermia (NOA) with 91% sensitivity. [\[25\]](#)
- **Limitations:** Most models require large, labeled datasets for training, and generalizability across diverse populations remains a challenge. [\[7\]](#)

8.3 Commercial Solutions in AI-Based Semen Analysis

The global semen analysis market is projected to grow at a 6.5–8% CAGR (2025–2032), driven by AI integration and rising infertility rates [\[31\]](#) [\[33\]](#) . Key players and innovations include:

- **Medical Electronic Systems:** SQA-V GOLD analyzer, validated for rapid, standardized results (95% correlation with manual methods). [\[30\]](#)
- **ExSeed Health:** Smartphone-connected kits for at-home motility and count analysis, enhancing accessibility. [\[31\]](#) [\[32\]](#)
- **Microptic:** AI-powered CASA systems with cloud integration for clinics, reducing turnaround time by 50%. [\[31\]](#) [\[33\]](#)
- **CooperSurgical:** FDA-approved AI tools for sperm selection in IVF, improving fertilization rates. [\[31\]](#) [\[32\]](#)
- **CooperSurgical:** Portable microfluidic devices (e.g., YO™ Sperm Test) and AI-driven DNA fragmentation tests (e.g., SpermQT). [\[7\]](#) [\[33\]](#)

9 Future Directions in AI-Driven Semen Analysis

Advancements in AI for semen analysis must address interdisciplinary challenges to transition from research labs to clinical practice. Here's a granular roadmap:

9.1 Integration with Emerging Technologies

9.1.1 IoT and Wearable Devices

Sensor-equipped devices (e.g., smart cups, microfluidic chips) could enable real-time sperm quality monitoring. For example, IoT-enabled microscopes paired with AI may reduce dependency on lab infrastructure ¹⁰. A 2023 prototype combining AI with a smartphone-based lens achieved 92% concordance with lab-grade motility analysis . [\[35\]](#)

9.1.2 Robotics

AI-driven robotic systems (e.g., automated sperm retrieval platforms) could automate sperm selection for ICSI, improving precision and IVF success rates.[\[36\]](#)

9.1.3 Blockchain

Secure, decentralized health records could facilitate anonymous data sharing between clinics while maintaining patient privacy.

9.2 Personalized Medicine

9.2.1 Predictive Analytics

AI algorithms (e.g., gradient boosting trees) could forecast individual sperm quality trends using hormonal, genetic, and lifestyle data. [\[25\]](#)

9.2.2 Treatment Optimization

Dose-response models for fertility drugs could be tailored to semen parameters and biomarkers. A 2023 trial achieved 91% sensitivity in predicting sperm retrieval success using AI. [\[25\]](#)

9.3 Larger and Diverse Datasets

9.3.1 Current Limitations

- **Bias:** A 2024 audit found 78% of semen AI models were trained on data from European males, leading to 15–30% accuracy drops in African/Asian cohorts. [\[25\]](#)

9.3.2 Solutions

To address the bias introduced by regionally limited training datasets in semen analysis AI, federated learning (FL) offers a decentralized training strategy. Rather than sharing raw patient data across institutions, FL allows multiple clinical sites to collaboratively train AI models by exchanging only learned parameters. This preserves data privacy and complies with local data regulations. A notable example, although applied to oocyte competence prediction, involved 8 IVF clinics across 6 countries (USA, Japan, India, Malaysia, Turkey, Trinidad & Tobago), where a federated model was trained on 10,677 images without sharing sensitive data. The federated model achieved 83–88% sensitivity on blind datasets and showed up to a 14% improvement in predictive performance when confounding factors like male infertility were excluded. This international collaboration simulates the goals of a Global Male Fertility Atlas (GMFA), indicating FL's strong potential in future AI tools for male reproductive health. [\[39\]](#)

9.4 Explainable AI (XAI)

9.4.1 Techniques

- **Layer-Wise Relevance Propagation (LRP):**
 - Used in ResNet-50 models to highlight sperm head defects in morphology analysis. Clinicians reported 55% faster diagnostics with LRP heatmaps. [\[38\]](#)
- **SHAP Values:**
 - Quantified the impact of smoking (SHAP = +0.23) and obesity (SHAP = +0.18) on AI-predicted motility scores.

9.4.2 Clinical Impact

Clinicians are more likely to adopt AI tools with interpretable outputs, aligning with regulatory demands for transparency. [\[37\]](#)

9.5 Regulatory Approval

9.5.1 FDA Pathway

- **Phase 1 (Preclinical):** Validate AI on 5,000+ samples across \$3 ethnic groups.
- **Phase 2 (Clinical):** Prove non-inferiority to WHO manual methods in a 12-month multicenter trial.
- **Phase 3 (Post-Market):** Monitor real-world performance via blockchain audited logs.

10 Conclusion

10.1 Summary of Key Findings

- AI-driven techniques from classical machine learning classifiers (e.g., SVM, Random Forest) to modern deep learning architectures (CNNs, RNNs, Transformers) have demonstrated substantial gains in accuracy, objectivity, and throughput for sperm concentration, motility, morphology, and DNA fragmentation analysis.
- Public datasets (VISEM, SVIA, HuSHeM, SCIAN-MorphoSpermGS) combined with open-source and commercial tools (DeepSperm, SpermQ, CASA systems) have catalyzed model development and validation, though data scarcity and variability remain challenges.
- Novel advances such as generative data augmentation (GANs), multi modal models integrating clinical metadata, real-time tracking, and ensemble approaches have further pushed the envelope towards robust, generalizable AI solutions.

10.2 Potential Impact

- In clinical practice, AI can standardize semen analysis across laboratories, reducing inter-observer variability and accelerating diagnosis from hours to minutes, thereby optimizing patient throughput and lowering costs.
- Enhanced predictive models enable personalized fertility treatments by correlating sperm metrics with genetic, lifestyle, and biochemical factors, paving the way for tailored intervention strategies and improved assisted reproductive technology (ART) outcomes.
- Integration with emerging technologies IoT-enabled at-home testing, robotic sperm selection, and secure blockchain data sharing promises a seamless, patient-centric workflow from sample collection to treatment planning.

10.3 Call to Action

- Expand and diversify annotated datasets via federated learning initiatives and synthetic data generation to mitigate bias and boost model generalizability across global populations.

- Invest in explainable AI (XAI) methods (e.g., SHAP, LRP) and rigorous clinical trials to build clinician trust and fulfill regulatory requirements (FDA, CE) for AI-based semen analysis devices.
- Foster interdisciplinary collaboration among andrologists, AI researchers, engineers, and regulatory bodies to co-design user-friendly, ethically sound solutions that translate AI breakthroughs into routine reproductive health care.

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